



Supporting Information

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Supplementary material

Table S1: Definition of unwanted groups

Name	Smarts pattern
> 2 ester groups	<chem>C(=O)O[C,H1].C(=O)O[C,H1].C(=O)O[C,H1]</chem>
2-halo pyridine	<chem>n1c([F,Cl,Br,I])cccc1</chem>
acid halide	<chem>C(=O)[Cl,Br,I,F]</chem>
acyclic C=C-O	<chem>C=[C!r]O</chem>
acyl cyanide	<chem>N#CC(=O)</chem>
acyl hydrazine	<chem>C(=O)N[NH2]</chem>
aldehyde	<chem>[CH1](=O)</chem>
Aliphatic long chain	<chem>[R0;D2][R0;D2][R0;D2][R0;D2]</chem>
alkyl halide	<chem>[CX4][Cl,Br,I]</chem>
amidotetrazole	<chem>c1nnnn1C=O</chem>
aniline	<chem>c1cc([NH2])ccc1</chem>
azepane	<chem>[CH2R2]1N[CH2R2][CH2R2][CH2R2][CH2R2][CH2R2]1</chem>
Azido group	<chem>N=[N+]=[N-]</chem>
Azo group	<chem>N#N</chem>
azocane	<chem>[CH2R2]1N[CH2R2][CH2R2][CH2R2][CH2R2][CH2R2][CH2R2]1</chem>
benzidine	<chem>[cR2]1[cR2][cR2]([Nv3X3,Nv4X4])[cR2][cR2][cR2]1[cR2]2[cR2][cR2][cR2]([Nv3X3,Nv4X4])[cR2][cR2]2</chem>
beta-keto/anhydride	<chem>[C,c](=O)[CX4,CR0X3,O][C,c](=O)</chem>
biotin analogue	<chem>C12C(NC(N1)=O)CSC2</chem>
Carbo cation/anion	<chem>[C+,c+,C-,c-]</chem>
catechol	<chem>c1c([OH])c([OH,NH2,NH])ccc1</chem>
charged oxygen or sulfur atoms	<chem>[O+,o+,S+,s+]</chem>
quinone	<chem>C1(=[O,N])C=CC(=[O,N])C=C1</chem>
quinone	<chem>C1(=[O,N])C(=[O,N])C=CC=C1</chem>
conjugated nitrile group	<chem>C=[C!r]C#N</chem>
crown ether	<chem>[OR2,NR2]@[CR2]@[CR2]@[OR2,NR2]@[CR2]@[CR2]@[OR2,NR2]</chem>

Name	Smarts pattern
cumarine	<chem>c1ccc2c(c1)ccc(=O)o2</chem>
cyanamide	<chem>N[CH2]C#N</chem>
cyanate / aminonitrile / thiocyanate	<chem>[N,O,S]C#N</chem>
cyanohydrins	<chem>N#CC[OH]</chem>
cycloheptane	<chem>[CR2]1[CR2][CR2][CR2][CR2][CR2][CR2]1</chem>
cycloheptane	<chem>[CR2]1[CR2][CR2]cc[CR2][CR2]1</chem>
cyclooctane	<chem>[CR2]1[CR2][CR2][CR2][CR2][CR2][CR2][CR2]1</chem>
cyclooctane	<chem>[CR2]1[CR2][CR2]cc[CR2][CR2][CR2]1</chem>
diaminobenzene	<chem>[cR2]1[cR2]c([N+0X3R0,nX3R0])c([N+0X3R0,nX3R0])[cR2][cR2]1</chem>
diaminobenzene	<chem>[cR2]1[cR2]c([N+0X3R0,nX3R0])[cR2]c([N+0X3R0,nX3R0])[cR2]1</chem>
diaminobenzene	<chem>[cR2]1[cR2]c([N+0X3R0,nX3R0])[cR2][cR2]c1([N+0X3R0,nX3R0])</chem>
diazo group	<chem>[N!R]=[N!R]</chem>
diketo group	<chem>[C,c](=O)[C,c](=O)</chem>
disulphide	<chem>SS</chem>
enamine	<chem>[CX2R0][NX3R0]</chem>
ester of HOBT	<chem>C(=O)Onnn</chem>
four member lactones	<chem>C1(=O)OCC1</chem>
halogenated ring	<chem>c1cc([Cl,Br,I,F])cc([Cl,Br,I,F])c1[Cl,Br,I,F]</chem>
halogenated ring	<chem>c1ccc([Cl,Br,I,F])c([Cl,Br,I,F])c1[Cl,Br,I,F]</chem>
heavy metal	<chem>[Hg,Fe,As,Sb,Zn,Se,se,Te,B,Si]</chem>
het-C-het not in ring	<chem>[NX3R0,NX4R0,OR0,SX2R0][CX4][NX3R0,NX4R0,OR0,SX2R0]</chem>
hydantoin	<chem>C1NC(=O)NC(=O)1</chem>
hydrazine	<chem>N[NH2]</chem>
hydroquinone	<chem>[OH]c1ccc([OH,NH2,NH])cc1</chem>
hydroxamic acid	<chem>C(=O)N[OH]</chem>
imine	<chem>C=[N!R]</chem>
imine	<chem>N=[CR0][N,n,O,S]</chem>
iodine	<chem>I</chem>

Name	Smarts pattern
isocyanate	<chem>N=C=O</chem>
isolated alkene	<chem>[\$([CH2]),\$([CH][CX4]),\$(C([CX4])[CX4]))=\$([CH2]),\$([CH][CX4]),\$(C([CX4])[CX4]))]</chem>
ketene	<chem>C=C=O</chem>
methylidene-1,3-dithiole	<chem>S1C=CSC1=S</chem>
Michael acceptor	<chem>C=!@CC=[O,S]</chem>
Michael acceptor	<chem>[\$([CH]),\$(CC)]#CC(=O)[C,c]</chem>
Michael acceptor	<chem>[\$([CH]),\$(CC)]#CS(=O)(=O)[C,c]</chem>
Michael acceptor	<chem>C=C(C=O)C=O</chem>
Michael acceptor	<chem>[\$([CH]),\$(CC)]#CC(=O)O[C,c]</chem>
N oxide	<chem>[NX2,nX3][OX1]</chem>
N-acyl-2-amino-5-mercapto-1,3,4-thiadiazole	<chem>s1c(S)nnc1NC=O</chem>
N-C-halo	<chem>NC[F,Cl,Br,I]</chem>
N-halo	<chem>[NX3,NX4][F,Cl,Br,I]</chem>
N-hydroxyl pyridine	<chem>n[OH]</chem>
nitro group	<chem>[N+](=O)[O-]</chem>
N-nitroso	<chem>[#7]-N=O</chem>
oxime	<chem>[C,c]=N[OH]</chem>
oxime	<chem>[C,c]=NOC=O</chem>
Oxygen-nitrogen single bond	<chem>[OR0,NR0][OR0,NR0]</chem>
perfluorinated chain	<chem>[CX4](F)(F)[CX4](F)F</chem>
peroxide	<chem>OO</chem>
phenol ester	<chem>c1ccccc1OC(=O)[#6]</chem>
phenyl carbonate	<chem>c1ccccc1OC(=O)O</chem>
phosphor	<chem>P</chem>
phthalimide	<chem>[cR,CR]~C(=O)NC(=O)~[cR,CR]</chem>
Polycyclic aromatic hydrocarbon	<chem>a1aa2a3a(a1)A=AA=A3=AA=A2</chem>
Polycyclic aromatic hydrocarbon	<chem>a21aa3a(aa1aaaa2)aaaa3</chem>

Name	Smarts pattern
Polycyclic aromatic hydrocarbon	<chem>a31a(a2a(aa1)aaaa2)aaaa3</chem>
polyene	<chem>[CR0]=[CR0][CR0]=[CR0]</chem>
quaternary nitrogen	<chem>[s,S,c,C,n,N,o,O]~[nX3+,NX3+](~[s,S,c,C,n,N])~[s,S,c,C,n,N]</chem>
quaternary nitrogen	<chem>[s,S,c,C,n,N,o,O]~[n+,N+](~[s,S,c,C,n,N,o,O])(~[s,S,c,C,n,N,o,O])~[s,S,c,C,n,N,o,O]</chem>
quaternary nitrogen	<chem>[*]=[N+]=[*]</chem>
saponine derivative	<chem>O1CCCCC1OC2CCC3CCCCC3C2</chem>
silicon halogen	<chem>[Si][F,Cl,Br,I]</chem>
stilbene	<chem>c1cccc1C=Cc2cccc2</chem>
sulfinic acid	<chem>[SX3](=O)[O-,OH]</chem>
Sulfonic acid	<chem>[C,c]S(=O)(=O)O[C,c]</chem>
Sulfonic acid	<chem>S(=O)(=O)[O-,OH]</chem>
sulfonyl cyanide	<chem>S(=O)(=O)C#N</chem>
sulfur oxygen single bond	<chem>[SX2]O</chem>
sulphate	<chem>OS(=O)(=O)[O-]</chem>
sulphur nitrogen single bond	<chem>[SX2H0][N]</chem>
Thiobenzothiazole	<chem>c12cccc1(SC(S)=N2)</chem>
thiobenzothiazole	<chem>c12cccc1(SC(=S)N2)</chem>
Thiocarbonyl group	<chem>[C,c]=S</chem>
thioester	<chem>SC=O</chem>
thiol	<chem>[S-]</chem>
thiol	<chem>[SH]</chem>
Three-membered heterocycle	<chem>*1[O,S,N]*1</chem>
triflate	<chem>OS(=O)(=O)C(F)(F)F</chem>
triphenyl methyl-silyl	<chem>[SiR0,CR0](c1cccc1)(c2cccc2)(c3cccc3)</chem>
triple bond	<chem>C#C</chem>